Exam question

Case study (3 points): In this exercise, your task is to apply the Bad Actors Modeling strategy to the SmartCity solution described in the scenario below:

- Choose 3 different categories among the five in the Bad Actors Modeling strategy (money, politics, entertainment, ideas, self-interest).
- Then for each of the categories you have chosen: identify and describe one harmful action a bad actor could take and how it could negatively impact stakeholders (1-2 sentences for each). Actions must be different from each other.

Scenario: The Swiss company SmartCity wants to help local authorities to better manage city infrastructures and resources by deploying a solution based on the Internet of Things (IoT) - a network of objects with computing capabilities connected to the Internet through Wi-Fi or 5G. The solution proposed by SmartCity uses a range of connected devices to monitor waste-bins, traffic flow as well as weather conditions throughout a city. Level monitoring sensors on waste-bins allow waste collection services to optimize their routes. Cameras on major roads and intersections are connected to machine learning models to analyze real-time traffic data and dynamically adjust green light times. Using real-time weather data, indoor ventilation and temperature are automatically adjusted to keep building occupants comfortable while minimizing energy use. Weather data is also used in predictive analytics to identify health risks and aggregated information is communicated to local communities in a public app, for instance in case of air pollution or heatwave. Staff from the city authorities can monitor and control the system remotely from a centralized dashboard. SmartCity plans to roll out its solution in a number of European countries with large cities, including France, Germany, Italy and Spain.

Grading rubric

1 point per bad actor action (total = 3 points)

- 0.5 the action described corresponds to the category (check the cheatsheet)
- 0.5 the description makes clear how the action is harmful / negatively impacts stakeholders and is specific to SmartCity (i.e. adapted to the scenario)

Penalizations:

- -0.5 if two actions are too similar to each other
- -0.25 if negative impact is not sufficiently described